

SEMINAR ON EMERGING TECHNOLOGIES

ZERO TRUST ARCHITECTURE AND MODERN CYBERSECURITY

Never Trust • Always Verify

A modern security approach for users, devices, applications and data.

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Introduction to the Emerging Technology

WHAT IS ZERO TRUST?

Zero Trust Architecture (ZTA) is a cybersecurity model that removes implicit trust. Every access request is verified using identity, device, context and security policy before access is granted.

CORE PRINCIPLE

Never Trust, Always Verify. Being inside a corporate network does not automatically make a user or device trusted.

SECURITY FOCUS

Identity • Least Privilege • Device Security • Continuous Monitoring • Micro-Segmentation

KEY POINT 1

Cloud computing has reduced the importance of a fixed network perimeter.

KEY POINT 3

Stolen credentials can allow attackers to appear as legitimate users.

KEY POINT 5

Organizations need continuous verification instead of one-time trust.

KEY POINT 2

Remote work allows users and devices to connect from many locations and networks.

KEY POINT 4

Third-party services, mobile devices and connected systems increase the attack surface.

KEY POINT 6

Least-privilege access helps reduce the impact of compromised accounts.

Working Principle / Architecture

01 USER / DEVICE

Requests access

02 IDENTITY CHECK

Authenticate user

03 CONTEXT CHECK

Assess device & risk

04 POLICY ENGINE

Evaluate rules

05 ACCESS CONTROL

Allow minimum
access

06 CONTINUOUS MONITOR

Re-check activity

ACCESS DECISION = IDENTITY + DEVICE + CONTEXT + POLICY

PRINCIPLE

The user is not trusted simply because the request comes from inside the network.

Key Components and Technologies

1. IDENTITY & ACCESS

Authentication, authorization and role-based access.

2. MULTI-FACTOR

AUTHENTICATION

Adds verification beyond a password.

3. DEVICE SECURITY

Checks device health and compliance.

4. POLICY ENGINE

Makes access decisions using security rules.

5. LEAST PRIVILEGE

Gives only the permissions required.

6. MICRO-SEGMENTATION

Restricts movement between resources.

7. CONTINUOUS MONITORING

Detects unusual behavior and policy violations.

8. ENCRYPTION

Protects data in transit and at rest.

Applications / Real-World Use Cases

REMOTE WORKFORCE

Secure access for employees working outside the office.

CLOUD SECURITY

Protect cloud applications and distributed resources.

ENTERPRISE SYSTEMS

Control access to sensitive internal applications.

CRITICAL INFRASTRUCTURE

Strengthen access controls for high-value systems.

FINANCIAL SERVICES

Protect customer data and privileged operations.

HEALTHCARE

Control access to sensitive patient and clinical systems.

Advantages and Limitations

ADVANTAGES

- Reduces implicit trust
- Supports least-privilege access
- Limits unauthorized movement
- Improves visibility and monitoring
- Works well with cloud and remote environments
- Helps contain compromised accounts

LIMITATIONS

- Requires careful planning and policy design
- Legacy systems can be difficult to integrate
- Large environments create management complexity
- Continuous monitoring requires resources
- Poor configuration can affect user experience

Recent Developments and Trends

DEVELOPMENT 1

01

Identity-centric security is becoming more important as organizations move to cloud platforms.

DEVELOPMENT 2

02

Risk-based and adaptive access decisions can respond to changing user and device conditions.

DEVELOPMENT 3

03

Security platforms increasingly combine identity, endpoint, network and application signals.

DEVELOPMENT 4

04

Automation and analytics can improve detection and response to suspicious behavior.

DEVELOPMENT 5

05

Zero Trust principles are increasingly applied across hybrid and multi-cloud environments.

Challenges and Future Scope

CHALLENGES

- Migrating legacy applications
- Defining consistent policies
- Managing many identities and devices
- Maintaining visibility across distributed systems
- Balancing security and usability

FUTURE SCOPE

- More adaptive risk-based access
- AI-assisted security analytics
- Automated policy enforcement
- Stronger device and identity intelligence
- Wider adoption across cloud and IoT environments

Case Study / Example

REMOTE EMPLOYEE ACCESSING A COMPANY APPLICATION

01

ACCESS REQUEST

Employee requests access to a company application.

02

IDENTITY VERIFY

Username, password and MFA are checked.

03

DEVICE CHECK

Device security status and compliance are evaluated.

04

POLICY DECISION

Policy engine grants only the required permissions.

05

CONTINUOUS MONITORING

Activity is monitored after access is granted.

RESULT: ACCESS IS VERIFIED, LIMITED AND CONTINUOUSLY MONITORED.

CONCLUSION

Zero Trust Architecture replaces implicit trust with continuous verification. It focuses on identity, device security, least privilege, policy enforcement and monitoring.

This approach is well suited to modern cloud, remote and distributed environments where the traditional network boundary is no longer sufficient.

REFERENCES

- NIST SP 800-207 — Zero Trust Architecture
- CISA — Zero Trust Maturity Model
- NIST Cybersecurity Framework
- Microsoft Zero Trust guidance
- Industry cybersecurity architecture resources

KEY TAKEAWAY: VERIFY • LIMIT ACCESS • MONITOR CONTINUOUSLY

THANK YOU

QUESTIONS & DISCUSSION

Zero Trust Architecture and Modern Cybersecurity